Susan Ho

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EDUCATION

University of Waterloo — Waterloo, ON | Sep. 2025 – Present

Double Degree Honours Mathematics – Financial Analysis & Risk Management (Co-op), Professional Risk Management Specialization & Statistics

- FARMSA Stock Pitch Competition: 2nd Place 2026, 3rd Place 2025 — 12–18% predicted gains
- USAMO Qualifier 2024: ranked top 500 nationally in selective mathematics competition

TECHNICAL SKILLS

Languages: Python, SQL (MySQL, PostgreSQL), VBA, C#, JavaScript, C/C++, Go

Finance & Risk: Financial Modeling, Credit Analysis, Risk Assessment, Financial Statement Analysis, Anomaly Detection, Cohort Analysis, Budget Reconciliation, KPI Reporting

Data & Analytics: ETL, Power BI, Pandas, NumPy, Scikit-learn, Plotly, Seaborn, Excel (Advanced)

Backend & Systems: REST APIs, Docker, .NET 8, ONNX, Linux/Unix

Tools: Git, GitHub, Workday (exposure), HRIS data integration

EXPERIENCE

Manifold Data Mining Inc. — North York, ON | Apr. 2026 – Aug. 2026

Software QA & Data Analyst | **Python**, **SQL**, data governance, anomaly detection, risk modeling

- Audited high-volume financial and demographic datasets using **SQL** queries, filtering low-accuracy clusters to support real-time economic risk modeling.
- Standardized complex data dictionaries and executed end-to-end stress tests, establishing data governance protocols to minimize edge-case risk failures.
- Engineered automated anomaly detection protocols in **Python** to enforce data integrity across spatial databases, mitigating upstream geocoding errors.
- Collaborated with data engineering teams to validate pipeline outputs, ensuring spatial database accuracy for downstream risk and demographic models.

City of Markham — Markham, ON | Dec. 2024 – Jul. 2025

Finance & Data Analyst Intern | **Python**, **VBA**, **Power BI**, **SQL**, financial reporting

- Developed 20+ automated financial reports with **VBA** and **Python**, reducing manual analyst effort by 40% and reallocating capacity to higher-value tasks.
- Designed **Power BI** dashboards tracking performance and financial KPIs, reviewed weekly by senior management to support financial decision-making.
- Reconciled historical financial datasets and corrected $150K in financial discrepancies, strengthening data integrity across budgetary reporting systems.
- Analyzed budget variance trends and flagged financial risk indicators, supporting senior analysts in quarterly financial planning and review cycles.

Adobe — Remote | Dec. 2023 – Mar. 2024

Data Analyst Intern | **SQL**, **Python**, cohort analysis, revenue reporting

- Performed cohort analysis on 500K+ user sessions, surfacing retention drop-off patterns to inform product strategy and support revenue risk assessments.
- Built reusable **SQL** reporting datasets adopted across 3+ teams, streamlining cross-functional financial reporting and improving data consistency by 20%.
- Investigated revenue pipeline discrepancies and resolved reporting inconsistencies, restoring confidence in financial metrics across finance stakeholders.
- Presented cohort analysis findings to cross-functional stakeholders, translating data insight into actionable recommendations for product and finance teams.

PROJECTS

Futures Market Risk Analysis Model | **Python**, **SQL**, **Pandas**, **NumPy**, **Plotly**, **PostgreSQL** | Dec. 2025 – Present

- Constructed a real-time futures data pipeline ingesting 50K+ daily records via Yahoo Finance API into **PostgreSQL** for price and volume monitoring.
- Applied Z-score anomaly detection to identify 65% of volume spikes correlated with adverse price movements, surfacing actionable market risk signals.
- Produced automated risk monitoring dashboards using **Python** and **Plotly**, cutting daily futures review from 4 hours to 1 hour largely for risk analysts.
- Validated model outputs against historical price benchmarks, quantifying forecast accuracy and refining Z-score thresholds to reduce false positive signals.

Intelligent Flight Price Predictor | **Python**, **C#**, **ONNX**, **Docker**, **REST API**, **.NET 8** | Jan. 2026 – Present

- Trained a **Random Forest** regression model on 200K+ flight records, achieving a mean prediction error of $23/ticket for quantitative pricing analysis.
- Deployed the model as a **.NET 8 REST API** using **ONNX** runtime, delivering sub-100ms pricing inference for real-time quantitative decision support.
- Scheduled model retraining pipelines to refresh fare predictions as market conditions evolve, maintaining forecast accuracy for quantitative decisions.
- Assessed model performance using MAE and RMSE metrics, benchmarking predictions against baseline fare estimates to validate quantitative reliability.

Automated Financial Reporting System | **VBA**, **SQL**, **Excel**, **Power BI**, **Docker** | Oct. 2024 – Sep. 2025

- Processed 10K+ daily financial transactions and reduced report generation time from 6 hours to 30 minutes via automated **VBA** and **SQL** pipelines.
- Implemented 99%-precision anomaly detection to flag financial reporting errors, containerizing the full pipeline via **Docker** for production reliability.
- Visualized daily transaction trends and exception reports in **Power BI**, enabling finance teams to monitor reporting accuracy and flag discrepancies.
- Verified processed transaction outputs against source records daily, ensuring end-to-end financial data accuracy across automated reporting workflows.